Customer Shopping Behavior Analysis

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1 Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has noticed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline). They are particularly interested in uncovering which factors, such as discounts, reviews, seasons, or payment preferences, drive consumer decisions and repeat purchases.

The main business question addressed in this project is:

> *"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

2 Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across multiple product categories. The objective is to identify customer spending patterns, purchasing trends, customer segments, and subscription behavior to support data-driven business decisions.

The project combines:

- Python for data cleaning and preprocessing
- PostgreSQL for structured data analysis
- Power BI for dashboard visualization

3 Dataset Summary

The dataset contains customer demographic information, product details, and shopping behavior attributes.

Attribute	**Description**
Number of Rows	3,900
Number of Columns	18
Customer Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping Behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
Missing Data	37 missing values in Review Rating column

Table 1: Dataset Summary

4 Exploratory Data Analysis using Python

The exploratory data analysis and preprocessing were carried out using Python libraries such as pandas and NumPy.

4.1 Data Loading

The dataset was imported using pandas and stored in a DataFrame for analysis.

4.2 Initial Exploration

The following methods were used:

- `df.info()` to inspect dataset structure
- `df.describe()` for summary statistics

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

Figure 1: Dataset Information and Summary Statistics

4.3 Missing Data Handling

Missing values were identified in the `review_rating` column. These missing values were imputed using the median review rating for each product category.

4.4 Column Standardization

Column names were converted to snake case format to improve readability and maintain consistency across analysis.

4.5 Feature Engineering

Additional features were created to enhance analysis:

- `age_group` column by categorizing customers into age ranges
- `purchase_frequency_days` column based on purchase frequency data

4.6 Data Consistency Check

The columns `discount_applied` and `promo_code_used` were analyzed for redundancy. Since both represented similar information, the `promo_code_used` column was removed.

4.7 Database Integration

The cleaned dataset was connected to PostgreSQL using Python and uploaded into the database for SQL-based analysis.

5 Data Analysis using SQL

Structured SQL queries were executed in PostgreSQL to answer important business questions and identify trends.

5.1 Revenue by Gender

Compared total revenue generated by male and female customers.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

Figure 2: Revenue by Gender

5.2 High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Total rows: 839 Query complete 00:00:00.388

Figure 3: High-Spending Discount Users

5.3 Top 5 Products by Rating

Analyzed products with the highest average customer review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

Figure 4: Top 5 Products by Rating

5.4 Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping methods.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

Figure 5: Shipping Type Comparison

5.5 Subscribers vs Non-Subscribers

Compared average spending and total revenue across subscription categories.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

Figure 6: Subscribers vs Non-Subscribers

5.6 Discount-Dependent Products

Identified products with the highest percentage of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

Figure 7: Discount-Dependent Products

5.7 Customer Segmentation

Customers were classified into three segments:

- New Customers
- Returning Customers
- Loyal Customers

This segmentation was based on purchase frequency and previous purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returing	701

Figure 8: Customer Segmentation

5.8 Top 3 Products per Category

Identified the most frequently purchased products within each category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Figure 9: Top 3 Products per Category

5.9 Repeat Buyers and Subscription Behavior

Analyzed whether customers with more than five purchases were more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

Figure 10: Repeat Buyers and Subscription Analysis

5.10 Revenue by Age Group

Calculated the revenue contribution of different age groups.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

Figure 11: Revenue by Age Group

6 Dashboard in Power BI

An interactive Power BI dashboard was created to visualize customer purchasing behavior, revenue trends, customer segmentation, and product performance.

Figure 12: Power BI Dashboard

The dashboard includes:

- Revenue trends
- Customer segmentation analysis
- Product category performance
- Subscription insights
- Shipping and discount analysis

7 Business Recommendations

Based on the analysis, the following recommendations are proposed:

7.1 Boost Subscriptions

Promote exclusive benefits and offers for subscribers to improve customer retention and recurring revenue.

7.2 Customer Loyalty Programs

Reward repeat customers through loyalty programs to encourage movement into the loyal customer segment.

7.3 Review Discount Policies

Optimize discount strategies to balance increased sales with profit margin control.

7.4 Product Positioning

Highlight top-rated and best-selling products in marketing campaigns and product recommendations.

7.5 Targeted Marketing

Focus marketing campaigns on high-revenue age groups and customers preferring express shipping.

8 Conclusion

This project successfully analyzed customer shopping behavior using Python, PostgreSQL, and Power BI. The analysis identified important purchasing trends, customer segments, and revenue-driving factors.

The findings can help the company:

- Improve customer engagement

- Optimize marketing strategies
- Enhance product positioning
- Increase subscription conversion
- Improve long-term customer loyalty

The integration of data cleaning, SQL analysis, and dashboard visualization demonstrates a complete end-to-end data analytics workflow.